METRIC

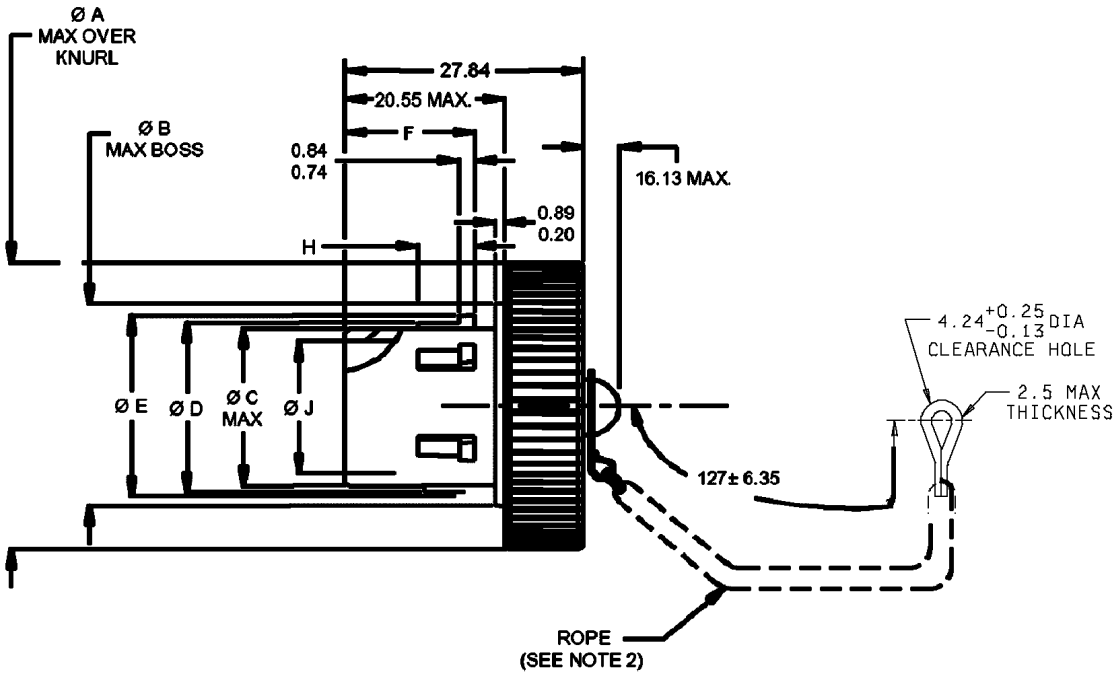
MIL-DTL-38999/51B
w/AMENDMENT 2
2 July 2024
SUPERSEDING
MIL-DTL-38999/51B
w/AMENDMENT 1
8 August 2017

DETAIL SPECIFICATION SHEET

CONNECTOR, ELECTRICAL, CIRCULAR, COVER, PROTECTIVE, PLUG, MIL-DTL-38999 SERIES IV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-38999.



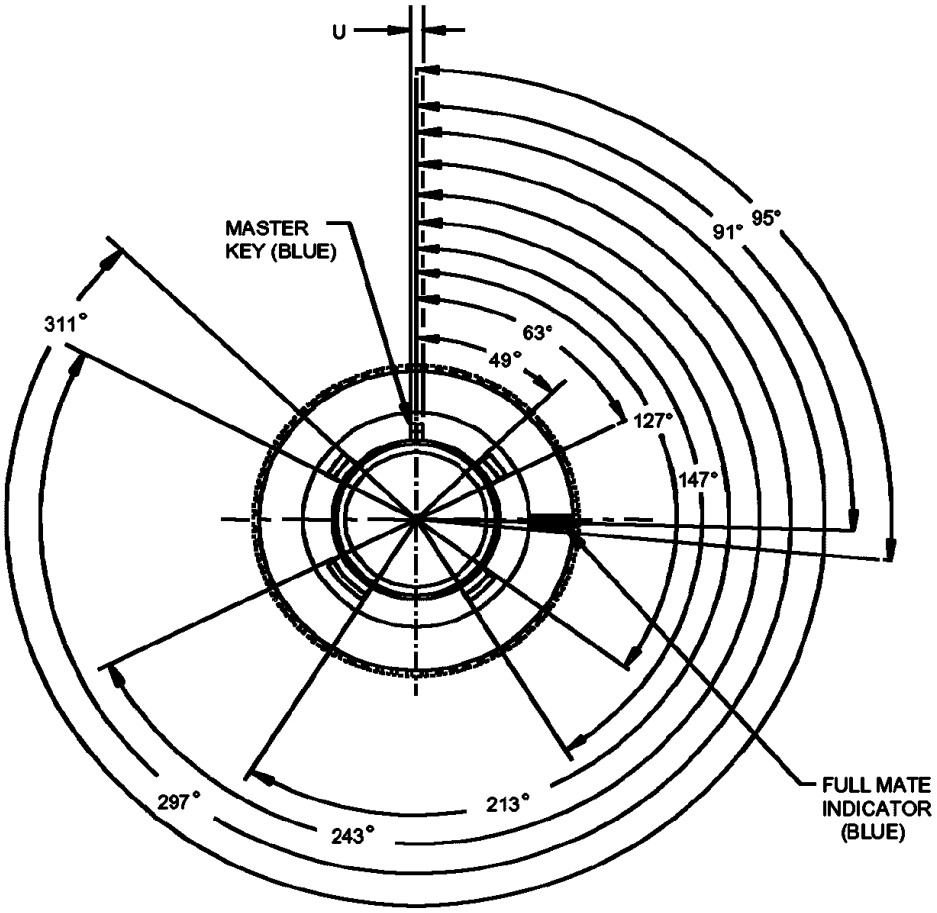
STYLE R (ROPE ONLY)

FIGURE 1. Protective cover, plug, all classes.

AMSC N/A FSC 5935

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mm	Inches	mm	Inches
0.20	.008	16.13	.635
0.74	.029	20.55	.809
0.84	.033	27.84	1.096
0.89	.035	127	5.000

FIGURE 1. Protective cover, plug, all classes - Continued.

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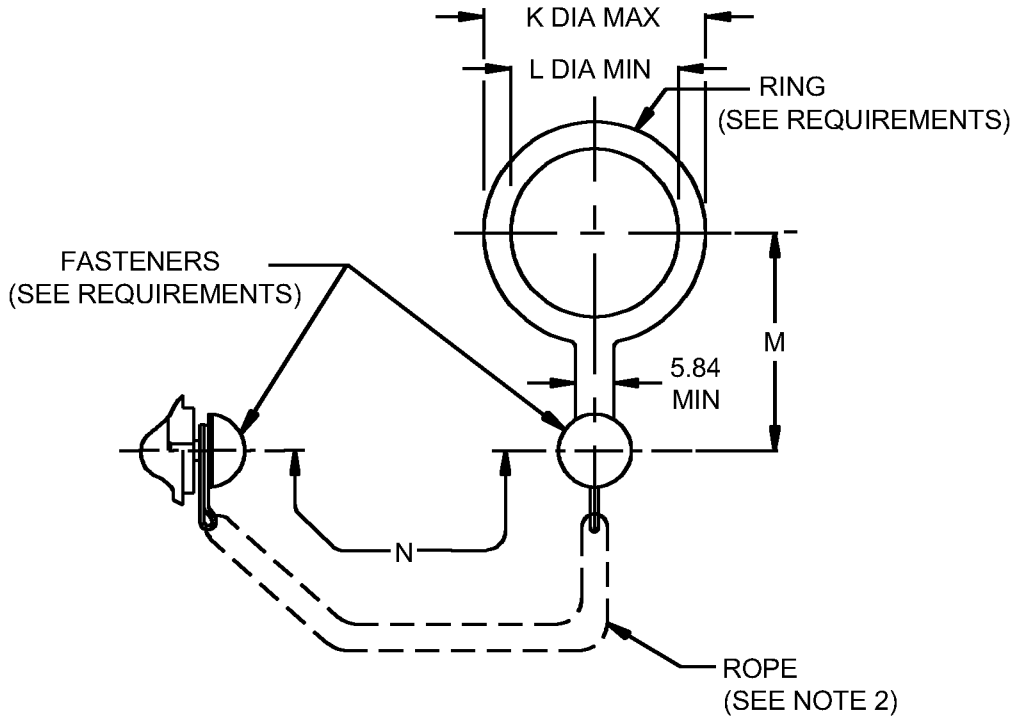
Dimensions									
Shell size	A max dia	B max dia	C dia	D dia	E dia	F	H	J dia min	U max
11	35.05 (1.380)	20.14 (.793)	12.929 (.509) 12.776 (.503)	14.986 (.590) 14.884 (.586)	16.26 (.640) 16.15 (.636)	17.145 (.675) 16.993 (.669)	7.010 (.276)	10.74 (.423)	1.651 (.065)
13	39.88 (1.570)	23.34 (.919)	16.10 (.634) 15.95 (.628)	18.313 (.721) 18.212 (.717)	19.33 (.761) 19.23 (.757)			13.64 (.537)	
15	42.17 (1.660)	26.52 (1.044)	19.28 (.759) 19.13 (.753)	21.463 (.845) 21.361 (.841)	22.48 (.885) 22.38 (.881)			16.81 (.662)	2.159 (.085)
17	49.47 (1.790)	29.72 (1.170)	22.48 (.885) 22.33 (.879)	24.638 (.970) 24.536 (.966)	25.65 (1.010) 25.55 (1.006)			19.99 (.787)	
19	48.77 (1.920)	32.87 (1.294)	25.63 (1.009) 25.48 (1.003)	27.81 (1.095) 27.76 (1.093)	27.66 (1.089) 27.56 (1.085)	16.891 (.665) 16.739 (.659)	6.756 (.266)	22.25 (.876)	2.667 (.105)
21	52.33 (2.060)	36.04 (1.419)	28.80 (1.134) 28.65 (1.128)	30.99 (1.220) 30.48 (1.200)	30.84 (1.214) 30.73 (1.210)			25.43 (1.001)	
23	57.15 (2.250)	39.22 (1.544)	31.98 (1.259) 31.83 (1.253)	34.29 (1.350) 33.78 (1.330)	34.11 (1.343) 34.01 (1.339)			28.60 (1.126)	3.175 (.125)
25	60.45 (2.380)	42.39 (1.669)	35.15 (1.384) 35.00 (1.378)	37.34 (1.470) 36.83 (1.450)	37.34 (1.470) 37.24 (1.466)			31.78 (1.251)	

NOTES:

1. Dimensions are in millimeters. Inch equivalents are given for information only.
2. Method of attachment to cover and or ring is optional.

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FIGURE 1. Protective cover, plug, all classes – Continued.



STYLE N (ROPE AND RING)

Dimensions				
Shell size	K dia max	L dia min	M	N
11	27.0 (1.063)	18.0 (.709)	20.50 (.807) 18.00 (.709)	127.00 (5.00)
13	30.0 (1.181)	20.0 (.788)	27.00 (1.06) 20.00 (.788)	127.00 (5.00)
15	31.0 (1.220)	23.0 (.907)	22.50 (0.886) 18.00 (.709)	127.00 (5.00)
17	37.0 (1.457)	26.0 (1.025)	30.50 (1.201) 23.50	127.00 (5.00)

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			(.925)	

FIGURE 2. Protective cover, plug, rope and ring assembly, all classes.

Shell size	K dia max	L dia min	M	N
19	40.0 (1.575)	29.0 (1.141)	31.50 (1.240) 25.00 (.984)	127.00 (5.00)
21	44.0 (1.732)	32.0 (1.260)	33.00 (1.30) 26.50 (1.043)	127.00 (5.00)
23	46.0 (1.811)	34.0 (1.339)	34.000 (1.339) 27.50 (1.083)	127.00 (5.00)
25	49.0 (1.929)	39.0 (1.538)	38.00 (1.496) 30.00 (1.181)	127.00 (5.00)

NOTES:
1. Dimensions are in millimeters. Inch equivalents are given for information only.
2. Method of attachment to cover and or ring is optional.

FIGURE 2. Protective cover, plug, rope and ring assembly, all classes – Continued.

REQUIREMENTS:

Dimensions and configurations: See figures 1 and 2.

Materials:

Cover - Classes C, F, G, R, T, W, and Z: Impact extruded or machined aluminum alloy.
Classes H, K, N, S and Y: Corrosion resistant steel.
Classes J and M: High performance resin in accordance with MIL-DTL-38999.
Class F: Impact extruded or machined aluminum alloy.
(Base materials listed. For finishes or surface treatments for each class, see MIL-DTL-38999.)

Gasket - Silicone elastomer.

Ring - Corrosion resistant steel, passivated, 1.020 mm (0.040 inches) maximum thickness.

Rope - Insulated stainless steel, passivated. Passivation or proof of passive finish using ASTM-A967/A967M or equivalent is permissible. Insulation shall withstand 200°C.

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environment.

Fastener - Stainless steel, passivated.
Option: Aluminum integral to cover.

Mating connectors: MIL-DTL-38999, series IV plugs.

Gasket shall be bonded to cover, or mechanically retained.

Rope shall rotate freely on fastener.

Fastener tensile strength: The protective cover and rope assembly shall withstand a dead weight tensile load of 25 pounds applied to both axial and longitudinal directions. The load shall be applied at the end of the rope assembly and held for 5 minutes. There shall be no separation of the rope assembly from the protective cover or damage to the rope assembly. Both axial and longitudinal directions shall be 25 pounds. The load shall be applied to the end of the rope.

Air Leakage: Testing shall be in accordance with EIA-364-02, except the minimum period of measurement shall be 10 seconds. The air leakage rate shall be no greater than 1 atmospheric cubic inch per hour (4.55×10^{-3} cm^3/s) Protective covers shall be mated to connectors having either contacts or inserts removed so that a pressure of 10 $lbft/in^2$ ($.703\ Kg/cm^2$) can be applied against the insides of the protective covers or stowage receptacles.

Qualification (all classes): Qualification shall be in accordance with MIL-DTL-38999 except only the following tests shall be required:

Examination of product
Coupling torque
Humidity
Salt spray (corrosion)
Fastener tensile strength (see requirement)
Air Leakage

Qualification (classes J and M only): Qualification shall be in accordance with MIL-DTL-38999, except that in addition to the test sequence specified for all classes above, the following tests shall be required:

Temperature cycling
Hydrolytic stability

Group A sampling inspection. Sampling for group A inspection shall be as specified in table I.

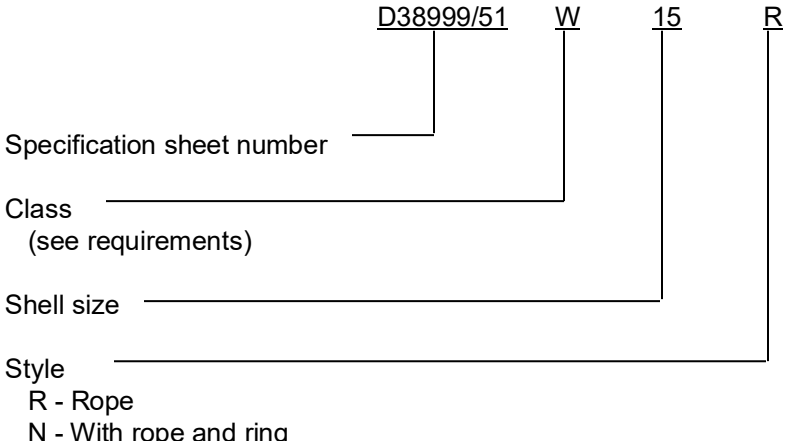
TABLE I. Group A sampling plan.

Lot size	Sampling size
1 to 13	100 percent
14 to 150	13 units
151 to 280	20 units
281 to 500	29 units
501 to 1,200	34 units
1,201 to 3,200	42 units

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3,201 and up	54 units

Part or Identifying Number (PIN) example:



Class F is not for Navy use and is inactive for use in Air Force new design.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-38999, this document references the following:

EIA-364-02
ASTM-A967/A967M

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2024-150)

Review activities:
Army - MI
Navy - EC, MC, OS
Air Force - 19

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